

1 REFLX: REACTION ENHANCEMENT IN FITNESS
2 USING LIGHT-BASED EXERCISES FOR UPV ATHLETES

3 A Special Problem
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5 the Faculty of the Division of Physical Sciences and Mathematics
6 College of Arts and Sciences
7 University of the Philippines Visayas
8 Miag-ao, Iloilo

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10 of the Requirements for the Degree of
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16 March 3, 2026

Approval Sheet

The Division of Physical Sciences and Mathematics, College of Arts and
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certifies that this is the approved version of the following special problem:

**REFLX: REACTION ENHANCEMENT IN FITNESS
USING LIGHT-BASED EXERCISES FOR UPV ATHLETES**

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Declaration

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Dedication

“Hello, world.”

Acknowledgment

“Hello, world.”

Abstract

From 150 to 200 words of short, direct and complete sentences, the abstract should be informative enough to serve as a substitute for reading the entire SP document itself. It states the rationale and the objectives of the research. In the final Special Problem document (i.e., the document you'll submit for your final defense), the abstract should also contain a description of your research results, findings, and contribution(s).

Suggested keywords based on ACM Computing Classification system can be found at https://dl.acm.org/ccs/ccs_flat.cfm

Keywords: Keyword 1, keyword 2, keyword 3, keyword 4, etc.

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Chapter 1

Introduction

1.1 Overview of the Current State of Technology

In the realm of sports science, the enhancement of athletes' response time and agility remains a critical focus, as these attributes directly influence performance in dynamic, unpredictable environments such as team sports and combat disciplines (Hassan et al., 2023). Traditional training methods, including cone drills and ladder exercises, have long been employed to improve these skills, yet they often fall short in replicating the rapid, stimulus-driven demands of real-game scenarios. Over the past decade, light-based reaction training systems—devices utilizing visual stimuli like LED lights to prompt immediate motor responses—have emerged as innovative tools to bridge this gap. Systems such as FITLIGHT, BlazePod, and XLiGHT have been critically analyzed for their design features, including sensor connectivity, battery life, and operational reliability, revealing strengths in portability and customization but limitations in diagnostic precision

86 and validity (Ezhov et al., 2021).

87 Empirical studies have demonstrated that these systems can significantly enhance
88 visual-motor coordination, reaction speed, and cognitive functions. For instance,
89 interventions using FITLIGHT in small-sided games have led to marked improve-
90 ments in harmonic abilities (e.g., rhythmization and responsiveness) and basic
91 skills like dribbling among young basketball players (Hassan et al., 2023). Simi-
92 larly, a 10-week FITLIGHT program improved reaction times and dribbling speeds
93 in female basketball athletes, with effect sizes indicating substantial neural adapta-
94 tions (Hassan, 2025). In motorsport contexts, light-based reactive agility training
95 has boosted selective attention, cognitive flexibility, and cardiorespiratory capac-
96 ity in car racing drivers (Horváth et al., 2022). A systematic review of visual
97 training interventions, including light board and stroboscopic methods, further
98 corroborates these benefits, reporting 5-27% reductions in reaction time across
99 various sports, with greater efficacy in elite and younger athletes (Jothi et al.,
100 2025). Reliability assessments of systems like BlazePod have also affirmed their
101 validity for measuring simple and complex reactions in mixed martial arts (MMA)
102 athletes, with high intraclass correlations supporting their use in training proto-
103 cols (Polechoński et al., 2024).

104 Despite these advancements, significant gaps persist in the literature. While light-
105 based systems show promise in controlled settings, their predictive value for field-
106 based reactive agility remains limited, as evidenced by weak correlations between
107 laboratory reaction speeds and on-field performance in soccer players (Broodryk
108 et al., 2025). This suggests a disconnect between isolated visual stimuli and the
109 multifaceted perceptual-cognitive demands of sports, highlighting the need for
110 more integrated, sport-specific designs. Moreover, comparative analyses under-

score inconsistencies in system performance, such as variable Bluetooth stability and sensor delays, which could undermine training reproducibility (Ezhov et al., 2021). These issues are largely tied to the wireless protocols used by commercial systems—typically Bluetooth or Wi-Fi—both of which introduce transmission overhead and variable latency, limiting their suitability for real-time reaction measurement. This technological gap highlights the need to explore emerging low-latency communication protocols.

1.2 Problem Statement

While traditional methods such as cone and ladder drills have been widely used to enhance reaction time and agility, they often fail to simulate the complex, stimulus-driven conditions of real-game scenarios (Hassan et al., 2023). Recent light-based reaction training systems have shown potential in improving visual-motor coordination and cognitive response (Horváth et al., 2022; Jothi et al., 2025). However, these systems remain limited by high costs, connectivity issues, and questionable transferability of laboratory-based improvements to on-field performance (Broodryk et al., 2025; Ezhov et al., 2021).

To address these limitations, this study bridges engineering innovation and applied sports performance research by incorporating ESP-NOW—a low-latency, connectionless wireless communication protocol developed by Espressif—in its development of a low-cost and customizable light-based reaction training system. By leveraging ESP-NOW’s novel capabilities, the system aims to eliminate the communication bottlenecks present in existing technologies and provide a more

temporally accurate, cost-efficient, and reliable training tool. If such limitations in current systems persist, athletes and coaches will continue to rely on devices that cannot fully simulate realistic gameplay conditions nor guarantee measurement precision. Thus, the integration and evaluation of ESP-NOW become central to developing a responsive and scientifically validated reaction training system.

1.3 Research Objectives

1.3.1 General Objective

The goal of this study is to develop and evaluate a light-based reaction training system that enhances the response time and agility of athletes, specifically in racket sports such as badminton, tennis, and table tennis.

1.3.2 Specific Objectives

Specifically, this study aims to:

1. design and construct a device equipped with infrared sensors, RGB lights, and speakers while integrating ESP-NOW for accurate motion detection and response measurement,
2. develop a software application that manages the device operations, records performance data, and functions both online and offline,

- 150 3. calibrate and test the system to ensure precision, responsiveness, and syn-
151 chronization between hardware and software components.
- 152 4. conduct experimental trials assessing the system's effectiveness in improving
153 athletes' response time and agility compared to traditional training methods,
154 and
- 155 5. evaluate the system's usability, functionality, and overall user satisfaction
156 based on feedback from athletes and coaches.

157 1.4 Scope and Limitations of the Research

158 This study focuses on the design, development, and short-term evaluation of a
159 programmable light-based reaction training system specifically tailored for racket
160 sports, including badminton, tennis, and table tennis. These sports were selected
161 because they demand rapid visual processing, anticipatory decision-making, and
162 fine motor control—abilities strongly linked to reaction time and agility.

163 Experimental trials are conducted in controlled indoor training environments, us-
164 ing drills that simulate racket-sport scenarios such as serve returns, directional
165 changes, and split-step reactions. The prototype system incorporates LED visual
166 cues and integrated speakers to deliver multimodal stimuli, allowing assessment
167 of both single and dual sensory response conditions.

168 The evaluation is limited to short-term performance outcomes, measuring pre-
169 post changes in reaction and movement response metrics following exposure to the
170 prototype system. Long-term effects, such as learning retention, in-game transfer,

171 or perceptual-cognitive adaptations, are beyond the scope of this research.

172 1.5 Significance of the Research

173 This research integrates engineering, computer science, and sports science by de-
174 veloping a light- and sound-based reaction training system that enhances athletes'
175 response time and agility through multimodal stimuli and data-driven feedback.
176 Through a technical lens, the study contributes to the computer science commu-
177 nity by implementing real-time sensor processing, audio-visual cue synchroniza-
178 tion, and a user-centered software design. The inclusion of both infrared sensors
179 and speakers allows for dynamic and varied training scenarios that engage multi-
180 ple sensory pathways, thereby improving cognitive-motor coordination. The use
181 of ESP-NOW as the primary wireless protocol for synchronizing training stations
182 is the central innovation this study. Unlike traditional Bluetooth-based systems,
183 ESP-NOW enables stable, connectionless, and ultra-low-latency communication,
184 thereby ensuring accurate temporal measurements essential for reaction-time re-
185 search. Its novel application into a light-based athletic training device positions
186 this study at the intersection of sports technology and modern IoT communication
187 design.

188 From a societal perspective, the system democratizes access to advanced reaction
189 training technologies by providing an affordable and portable tool suitable for ath-
190 letes, coaches, and educational institutions. It can also serve as a supplementary
191 device for rehabilitation programs that aim to improve motor control and sensory
192 processing. By combining technical innovation with accessibility, this research

193 promotes evidence-based athletic development and supports the broader integra-
194 tion of smart, adaptive technologies in sports and human performance training.

Chapter 2

Review of Related Literature

2.1 Physiological Basis of Reaction and Agility

Reaction time and agility in athletes are influenced by a complex interplay of neurophysiological factors, including perceptual-cognitive processing, neural pathways, and motor control. Reaction time encompasses the interval from stimulus detection to response initiation, modulated by sensory input (primarily visual), central nervous system processing, and efferent motor commands. Agility, defined as the ability to change direction rapidly while maintaining balance and speed, integrates these with biomechanical elements like strength and coordination (Pojskic et al., 2019). Neurophysiological mechanisms involve the visual cortex for stimulus detection, the prefrontal cortex for decision-making, and the basal ganglia for motor planning, with reaction accuracy linked to efficient attentional orienting and split attention across multiple stimuli (Chow et al., 2022). The prefrontal studies highlight that perceptual factors, such as anticipation and visual search efficiency,

210 contribute significantly, alongside physical attributes like explosive strength and
211 elastic strength (Yildiz et al., 2020). For instance, factor analyses reveal indepen-
212 dent components: explosive strength for acceleration, elastic strength for rebound,
213 change-of-direction speed (CODS), and maximal strength. Correlations show that
214 faster reaction times predict superior agility test performances, such as in the Illi-
215 nois Agility Test or 20-m shuttle sprint, emphasizing the role of neural efficiency
216 (Wang et al., 2024).

217 Methodologies in these studies often employ systematic reviews and correlational
218 analyses, drawing from diverse athletic populations (e.g., team sports like soc-
219 cer) (Turna, 2020). Limitations include reliance on lab-based tests that may not
220 fully replicate field conditions, potentially underestimating contextual factors like
221 fatigue or cognitive load (Pojskic et al., 2019). Implications suggest that train-
222 ing targeting neuroplasticity—through repeated stimuli—can enhance synaptic
223 efficiency, reducing reaction latencies and improving agility in sports requiring
224 rapid responses, such as basketball or racing (Chow et al., 2022). Patterns indi-
225 cate stronger associations in high-level athletes, where cognitive fatigue and sleep
226 quality positively correlate with prolonged reaction times, highlighting the need
227 for holistic training approaches (Yildiz et al., 2020).

228 2.2 Existing Reaction Training Technologies

229 Commercial systems like FITLIGHT Trainer and BlazePod represent established
230 light-based technologies for reaction training, with evidence supporting their ef-
231 fectiveness in enhancing athletic performance. FITLIGHT, a wireless LED sensor

232 system, simulates game-like conditions to improve reaction time, reflexes, and
233 cognitive functions (Hassan, 2025). Studies demonstrate its reliability, with a
234 test-retest intraclass correlation coefficient (ICC) of 0.89 and minimal detectable
235 changes in reaction metrics (Myers, 2021). In young basketball players, a 10-week
236 FITLIGHT intervention improved executive functions (e.g., inhibition, working
237 memory) and fitness, though gains were comparable to traditional training, sug-
238 gesting added cognitive demand without superior outcomes (Hassan, 2025). An-
239 other trial integrated FITLIGHT into small-sided games for 18 weeks, yielding
240 significant enhancements in coordinative abilities and basic skills, outperforming
241 controls with large effect sizes (Steff et al., 2024).

242 BlazePod, a pod-based visual-cognitive system, shows similar promise, with ex-
243 cellent reliability in balance activities (Çekok & Anaforoğlu, 2025). A 6-month
244 program in adolescent soccer players improved simple reaction time and cognitive
245 tasks, but between-group differences were non-significant compared to standard
246 training, indicating contextual benefits rather than inherent superiority (Theofilou
247 et al., 2022). Methodologies typically use randomized controlled trials (RCTs)
248 with pre-post assessments, validated via tools like the Stroop Test for cognition
249 and TUG for effort (Çekok & Anaforoğlu, 2025). Limitations include fairly small
250 samples ($n = 20-70$) and short durations (6-18 weeks), risking overestimation
251 of effects due to novelty; discrepancies arise in massed vs. distributed schedul-
252 ing, with longer protocols showing sustained gains. Implications underscore these
253 systems' role in individualized drills, boosting engagement through gamification,
254 though high costs and proprietary designs limit accessibility for broader athletic
255 populations (Hassan, 2025).

2.3 Light- and Sensor-Based Training Research

Research on light-based stimuli consistently shows enhancements in sports performance, particularly cognitive-motor integration and agility. A 6-week RCT using Witty SEM lights for car racing drivers improved cognitive abilities and cardiorespiratory fitness, with significant group-time interactions (Horváth et al., 2022). In basketball, light stimulation exercises enhanced attention focus (visual/auditory) and skilled hand speed, with pre-post improvements attributed to neuroplastic adaptations (Shimi et al., 2025). VR-adapted light tasks revealed attentional mechanisms, with orienting speed predicting performance, emphasizing split attention for larger stimuli arrays (Shimi et al., 2025). Soccer-specific lighting interventions (6 months) manipulated visual processing, reducing reaction times under varied conditions, though transfer to matches was inferred rather than directly measured (Theofilou et al., 2022).

Methodologies favor RCTs with tools like the Vienna Test System for cognition and breath-by-breath gas analysis for physiology, ensuring objective evidence (Horváth et al., 2022). Limitations encompass single-blinding, uncontrolled nutrition, and lab-based focus, potentially inflating effects; patterns show greater benefits in open-skill sports, with discrepancies in exercise vs. rest conditions (Zhao et al., 2024). Implications highlight light stimuli's potential for dual physical-cognitive gains, transferable to agility-dependent sports, but call for ecological validity in future studies (Shimi et al., 2025).

2.4 Wireless Communication Protocols in IoT

The rapid evolution of Internet of Things (IoT) and embedded-device systems has introduced novel communication protocols designed for low-latency, low-power, peer-to-peer connectivity. Among these, ESP-NOW stands out as a connectionless, peer-to-peer protocol by Espressif that operates on the Wi-Fi PHY layer while bypassing the full TCP/IP stack, which reduces handshake overhead and enables low-latency broadcast and one-to-many messaging for ESP32 and ESP8266 devices (Becker et al., 2025). Multiple empirical evaluations have assessed ESP-NOW's performance under different environmental and traffic conditions.

An outdoor performance study reported one-way transmission delays as low as approximately 1–2 ms even at extended distances, showing that ESP-NOW maintains stable latency and packet delivery across open-field environments (Becker et al., 2025). A lab analysis further confirmed these results by comparing ESP-NOW directly with Bluetooth Low Energy (BLE), finding that ESP-NOW exhibited measurably faster transmission times, significantly lower jitter, and more stable multi-node communication—attributes essential for synchronized sensor systems (Labib et al., 2021). Similarly, a comparative evaluation showed that ESP-NOW outperformed both classic Bluetooth and standard Wi-Fi in key metrics such as transmission speed, barrier resistance, and latency, including tests through walls and indoor obstacles (Eridani et al., 2021). An indoor performance evaluation of ESP-NOW reported high packet-delivery ratios and consistent millisecond-level timing across typical indoor IoT deployment ranges (Urazayev et al., 2023). These studies show that ESP-NOW reliably achieves one-way transmission times in the range of about 1–5 ms with low jitter—performance characteristics.

2.5 Chapter Summary

Should include a table of related studies comparing them based on several criteria.

Highlight research gaps and the research problem.

The literature reveals notable gaps, particularly the scarcity of open-source, customizable systems for individualized athlete training. While commercial tools like FITLIGHT dominate, they are proprietary and costly, limiting adaptation for diverse needs (Seçkin et al., 2023). Emerging technologies emphasize open-source platforms for tele-exercise, enabling collaborative customization via AI and IoT, yet only 9% of trials target healthy populations, with inadequate standardization of training parameters (Rebelo et al., 2023). Gaps include high dropout in asynchronous modes, data security concerns, and underexplored adherence factors (Rebelo et al., 2023). Methodologies in gap analyses rely on narrative reviews, highlighting sustainability challenges in open-source projects (Seçkin et al., 2023). This implicates the need for accessible systems to democratize training, addressing disparities in research on healthy athletes.

Another notable gap is the absence of research exploring low-latency wireless communication protocols—particularly ESP-NOW—in the design of reaction-training technologies. Existing systems largely depend on Bluetooth, which suffers from variable transmission delays and connectivity interruptions. No current studies assess whether protocols like ESP-NOW can enhance timing precision or improve synchronization in multi-station athletic training systems, underscoring a significant technological gap this research intends to address.

Chapter 3

Research Methodology

This chapter lists and discusses the specific steps and activities that were performed to accomplish the project. The discussion covers the activities from pre-proposal to Final SP Writing.

3.1 Research Activities

Research activities include inquiry, survey, research, brainstorming, canvassing, consultation, review, interview, observe, experiment, design, test, document, etc. Be sure that for each method, process, or algorithm used, there is a justification why that method was chosen. The methodology also includes the following information:

- who is responsible for the task
- the resource person to be contacted

- 336 • what were done
- 337 • when and how long the activity was done
- 338 • where it was done
- 339 • why should the activity was done

340 Chapter 4

341 Results and Discussions

342 This chapter presents the results or the system of your SP. Include screenshots,
343 tables, or graphs and provide the discussion of results.

344 Chapter 5

345 Conclusion

346 This chapter summarizes your SP and provides conclusions regarding your results
347 and analyses. Provide recommendations on what ought to be done with your SP
348 or provide further directions on the topic you covered.

Chapter 6

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442 **Appendix A**

443 **Code Snippets**

444 Appendix B

445 Resource Persons

446 **Dr. Firstname1 Lastname1**

447 Role1

448 Affiliation1

449 emailaddr@domain.com

450 **Mr. Firstname2 Lastname2**

451 Role2

452 Affiliation2

453 emailaddr2@domain.com

454 **Ms. Firstname3 Lastname3**

455 Role3

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458